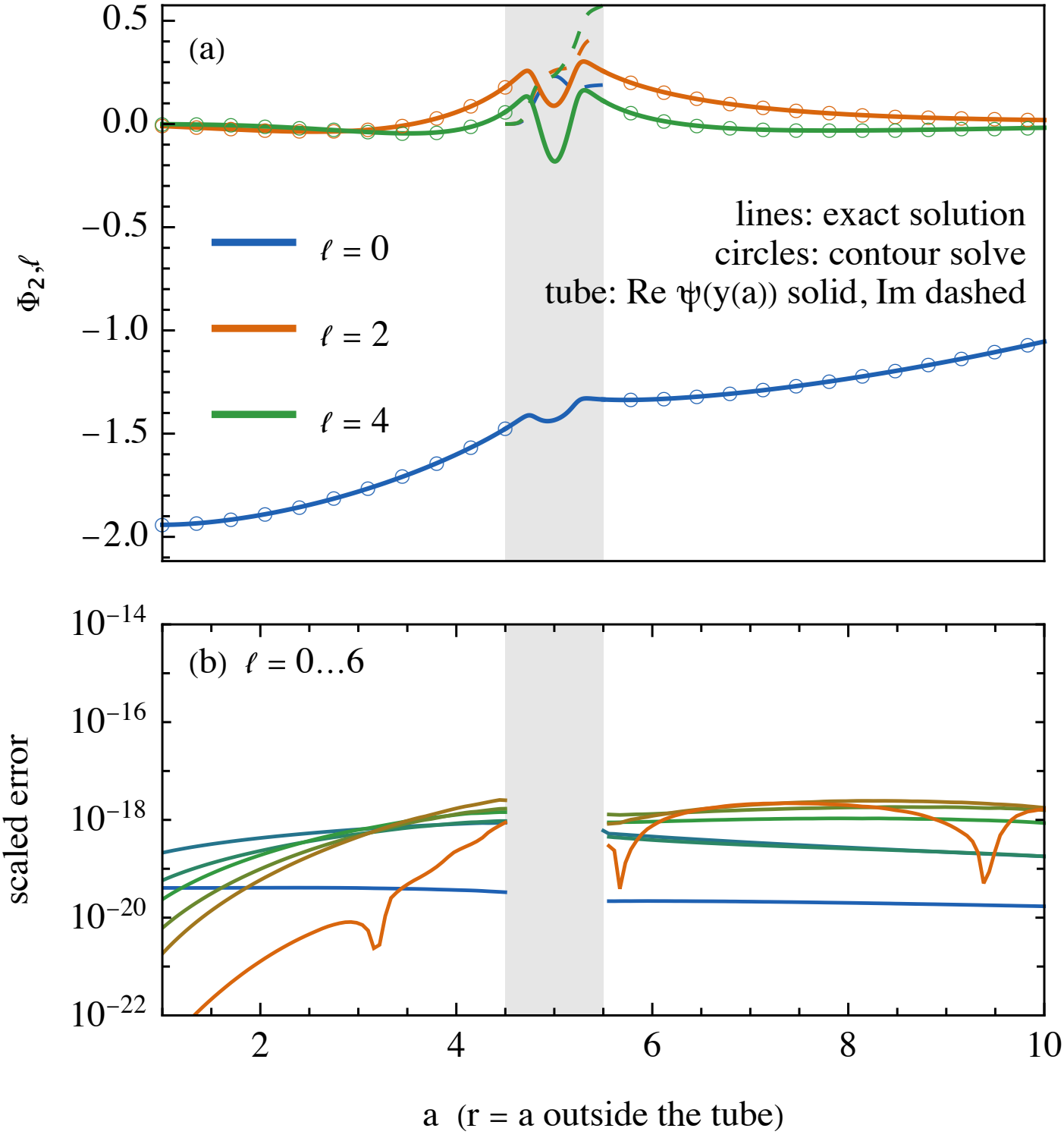
(a)
0.5
0.0
−0.5
−1.0
−1.5
−2.0
$\Phi_{2,\ell}$
$\ell = 0$
$\ell = 2$
$\ell = 4$
lines: exact solution
circles: contour solve
tube: Re ψ(y(a)) solid, Im dashed
(b) $\ell = 0\ldots6$
10^{-14}
10^{-16}
10^{-18}
10^{-20}
10^{-22}
scaled error
2
4
6
8
10
a (r = a outside the tube)